

AGORA

A few notes on the isolated disk comparison

Santa Cruz, 16 August, 2014

THE INITIAL CONDITIONS

Available at

<https://sites.google.com/site/projectagoraworkspace/collaborative-documents/isolated-disk-sims>

- Milky Way like disk setup using Volker Springel's **makedisk**
- Dark matter halo, stellar disk, stellar bulge and gas disk
- Low res (10^5 particles/species)
- Medium res (10^6 particles/species)
- High res (10^7 particles/species)

README.txt

UNITS

Velocity: km/s
Mass: 10^9 Msun
Length: kpc

Disk properties (in above units)

Disk scale length: 3.43218
Disk scale height: 0.343218 (10% of scale length)
M_DISK: 42.9661
M_GAS: 8.59322 (i.e. 20 % gas fraction)
M_200: 1074.15
R_200: 205.479
Total mass: 1301.7

Halo

c=10
v_200=150 km/s
lambda=0.04

For other quantities, see main.c

Number of particles and particle mass resolution

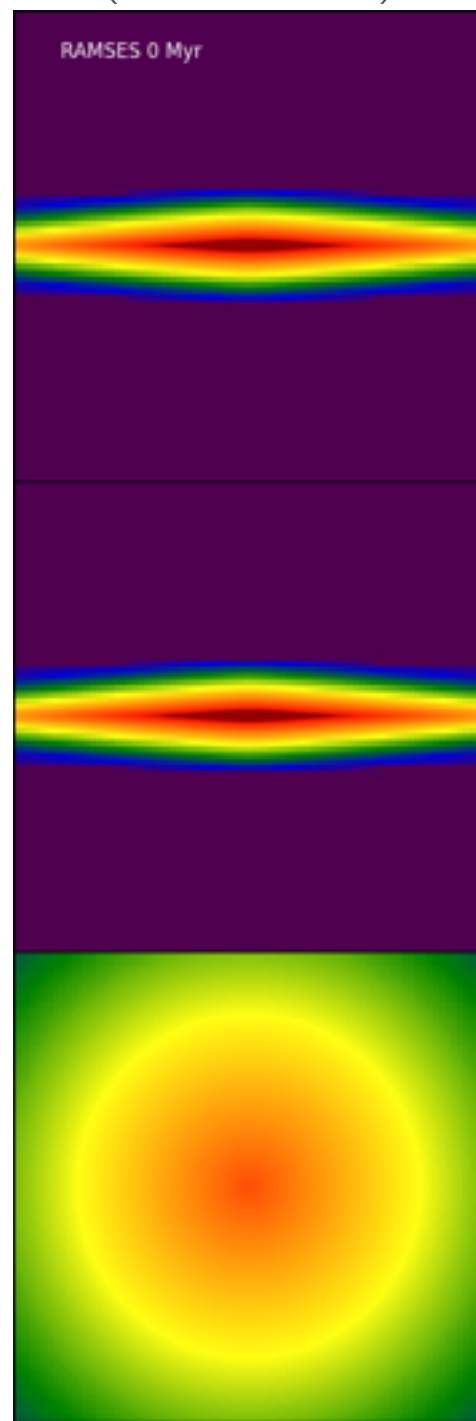
LOW

NGAS=1e5 ----> mgas_res = 8.593e4 Msun
NHALO=1e5 ----> mdark_res = 1.254e7 Msun
NDISK=1e5 ----> mdisk_res = 3.4373e5 Msun
NBULGE=1.25e4 ----> mbulge_res = 3.4373e5 Msun
(Bulge to total disk ratio=0.1, Bulge to stellar
disk=0.125)

THE INITIAL CONDITIONS

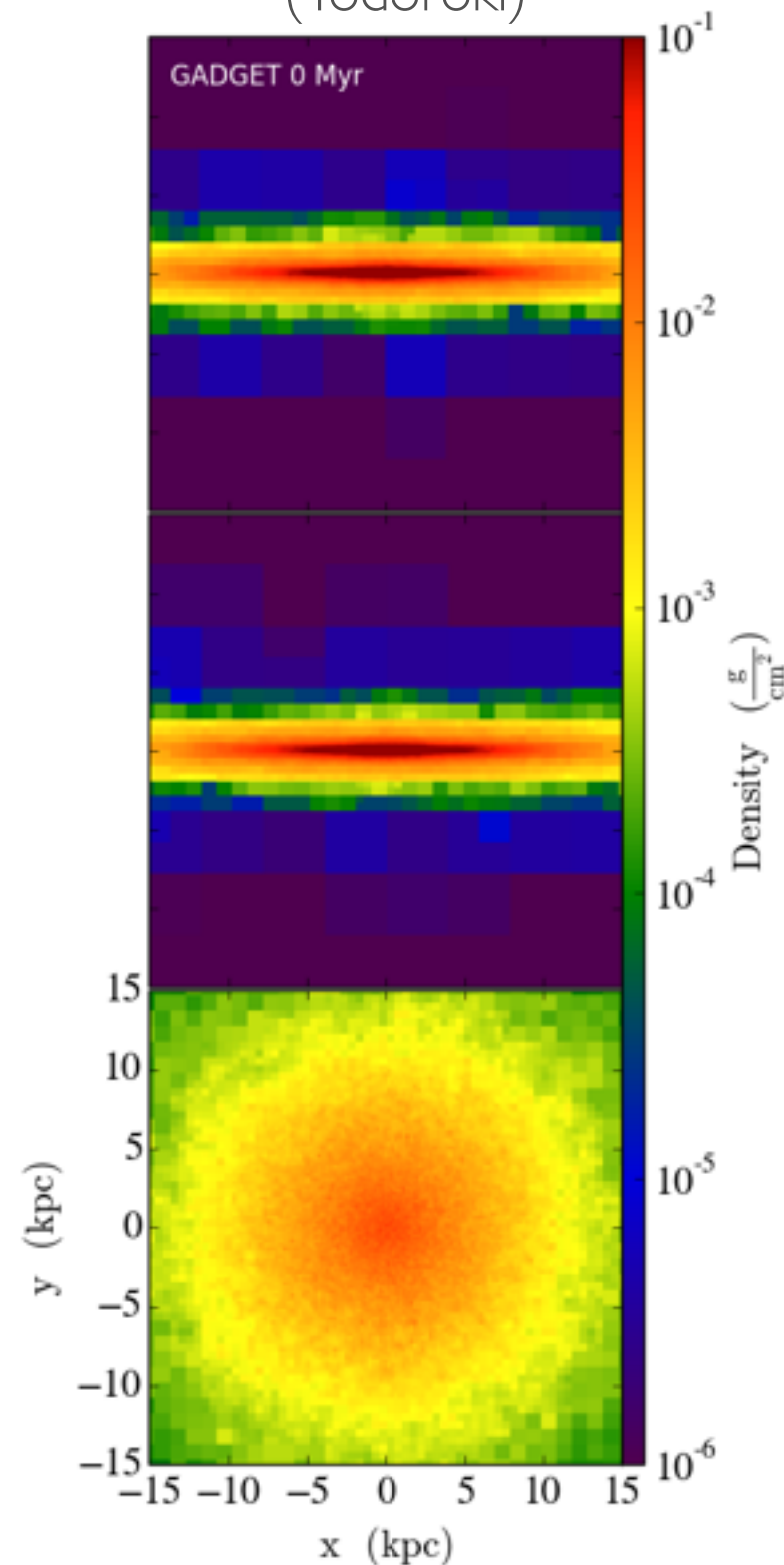
RAMSES

(Mike Butler)



Gadget-3

(Todoroki)

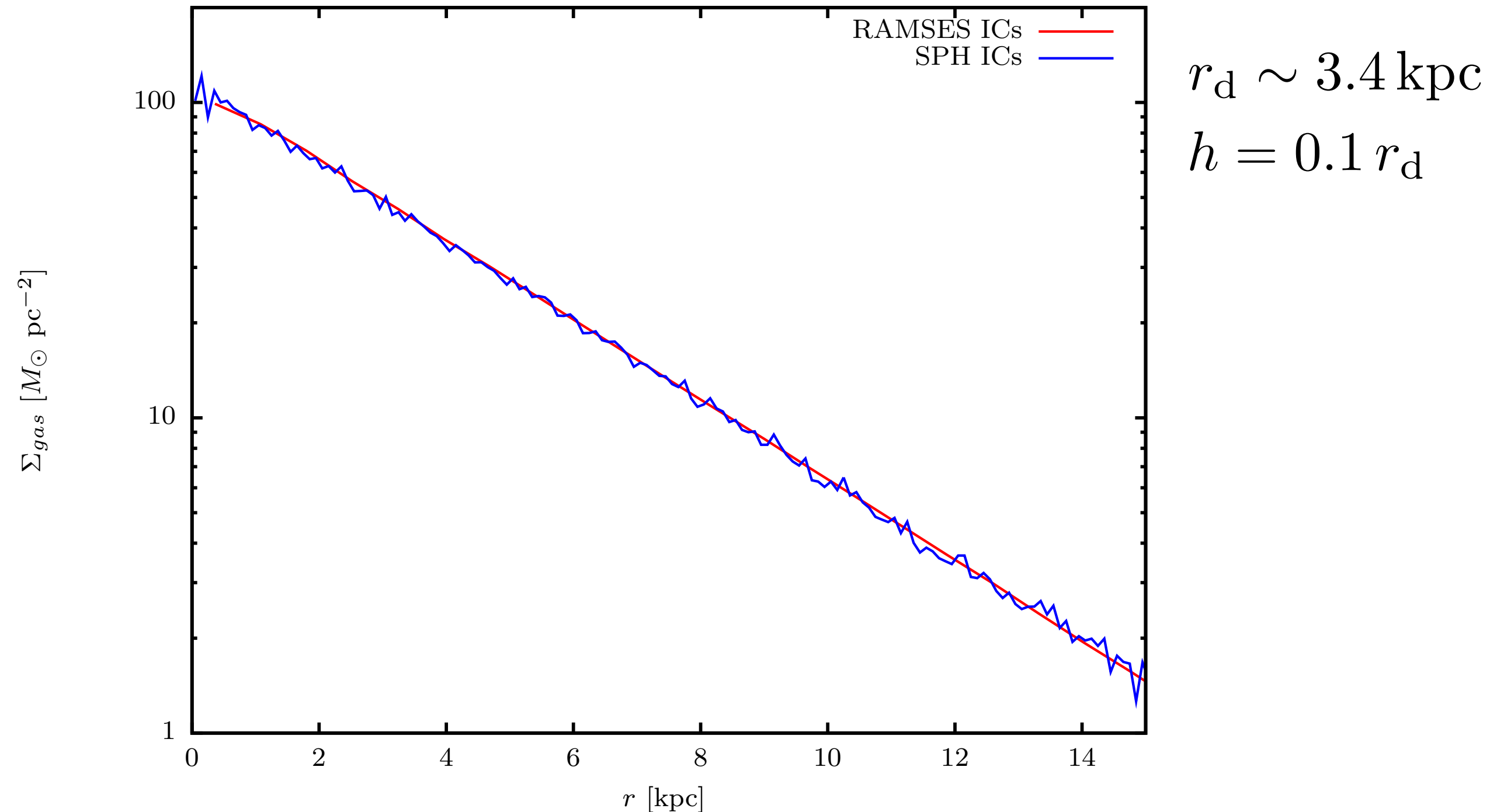


- **SPH:** Use supplied particle files for the dark matter halo, stellar disk, stellar bulge and gas disk
- **AMR:** Use the same collisionless particles, but use analytic profiles for the gas density and rotational profiles (i.e. “grid the gas”)

THE INITIAL CONDITIONS

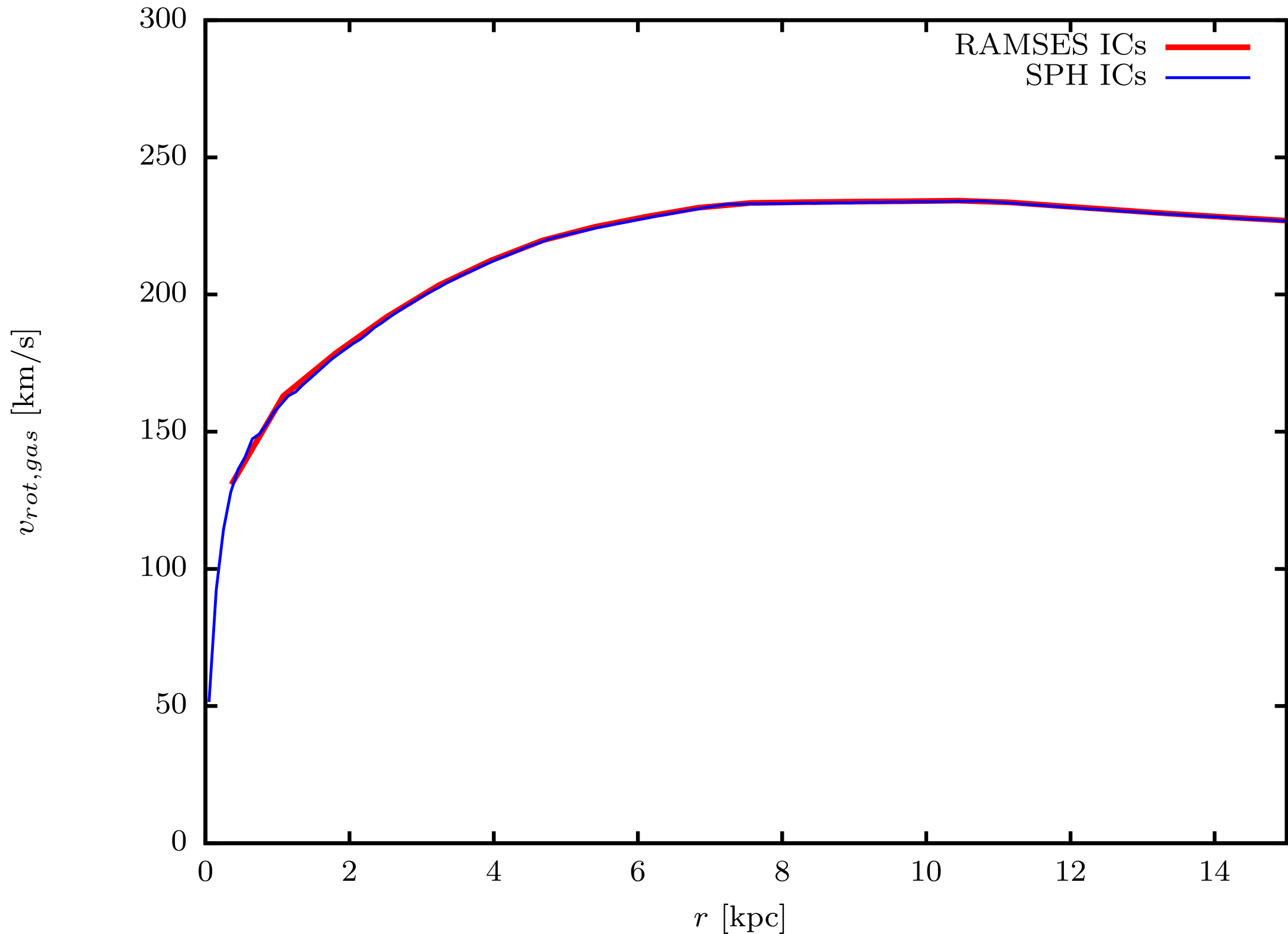
Gas density profile, sanity check

$$\rho(r, z) = \rho_0 \exp(-r/r_d) \exp(-z/h)$$



THE INITIAL CONDITIONS

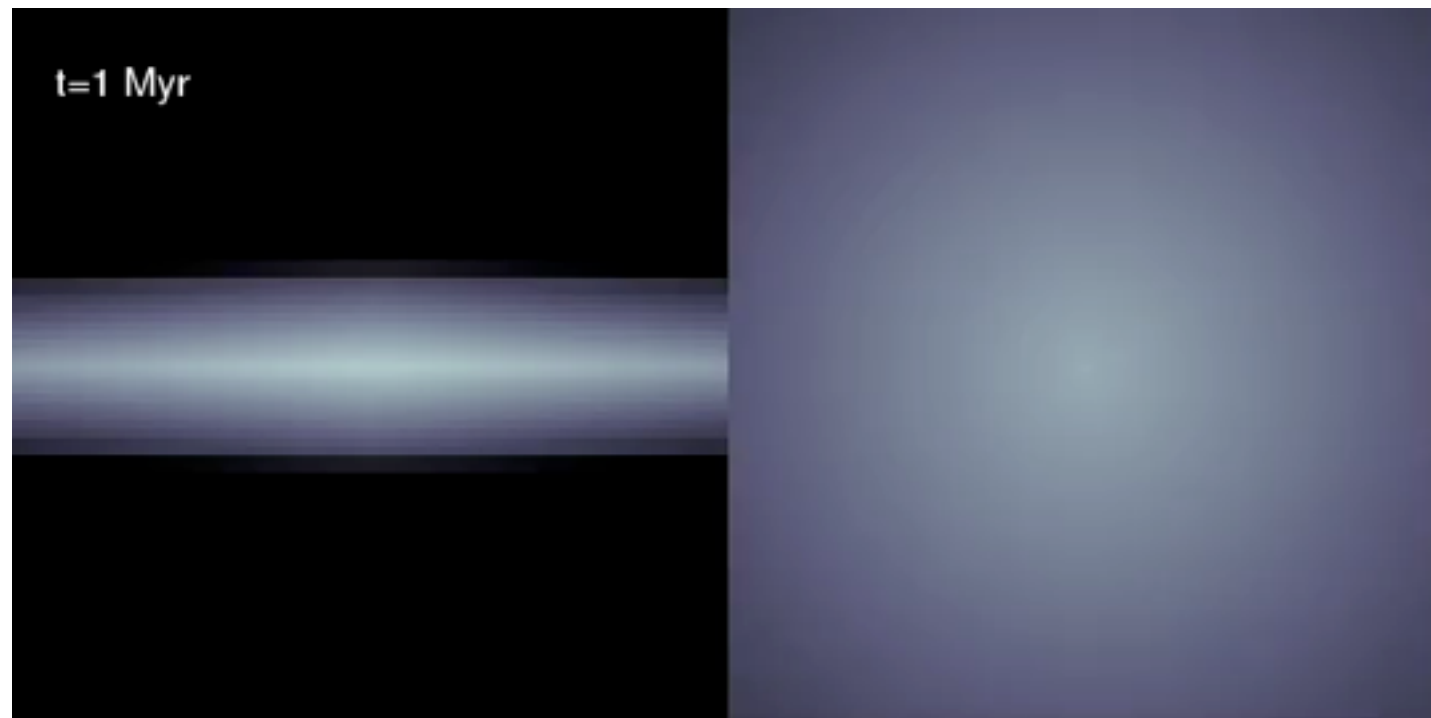
Gas rotational velocity profile, sanity check



A FEW EXAMPLES

RAMSES EXAMPLES

Low res, cooling + star formation (no feedback)

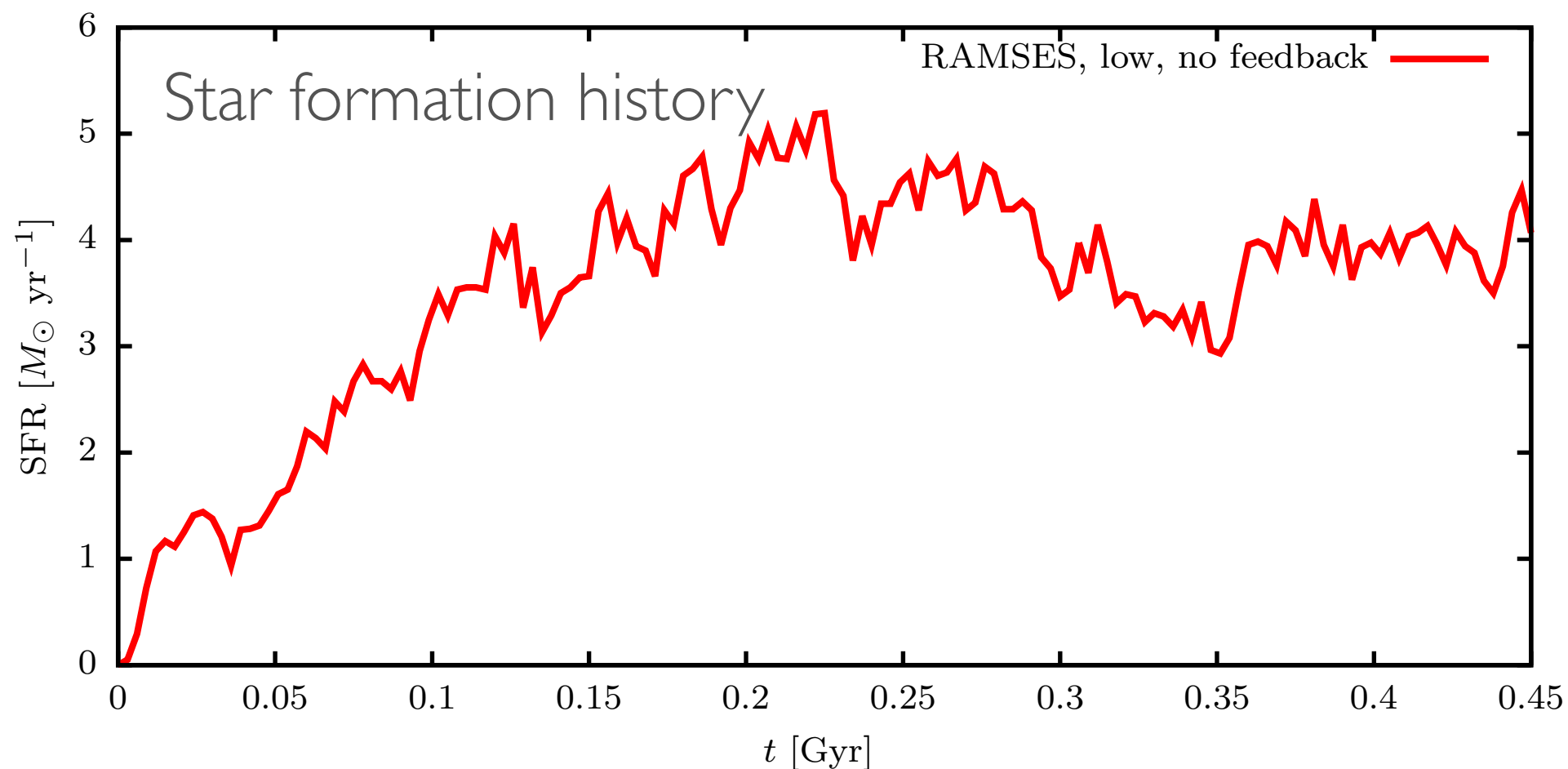


$$\dot{\rho}_* = \epsilon_{\text{ff}} \frac{\rho_{\text{gas}}}{t_{\text{ff}}}$$

$$\rho_{\text{threshold}} = 25 m_{\text{H}} \text{ cm}^{-3}$$

$$\epsilon_{\text{ff}} = 1\%$$

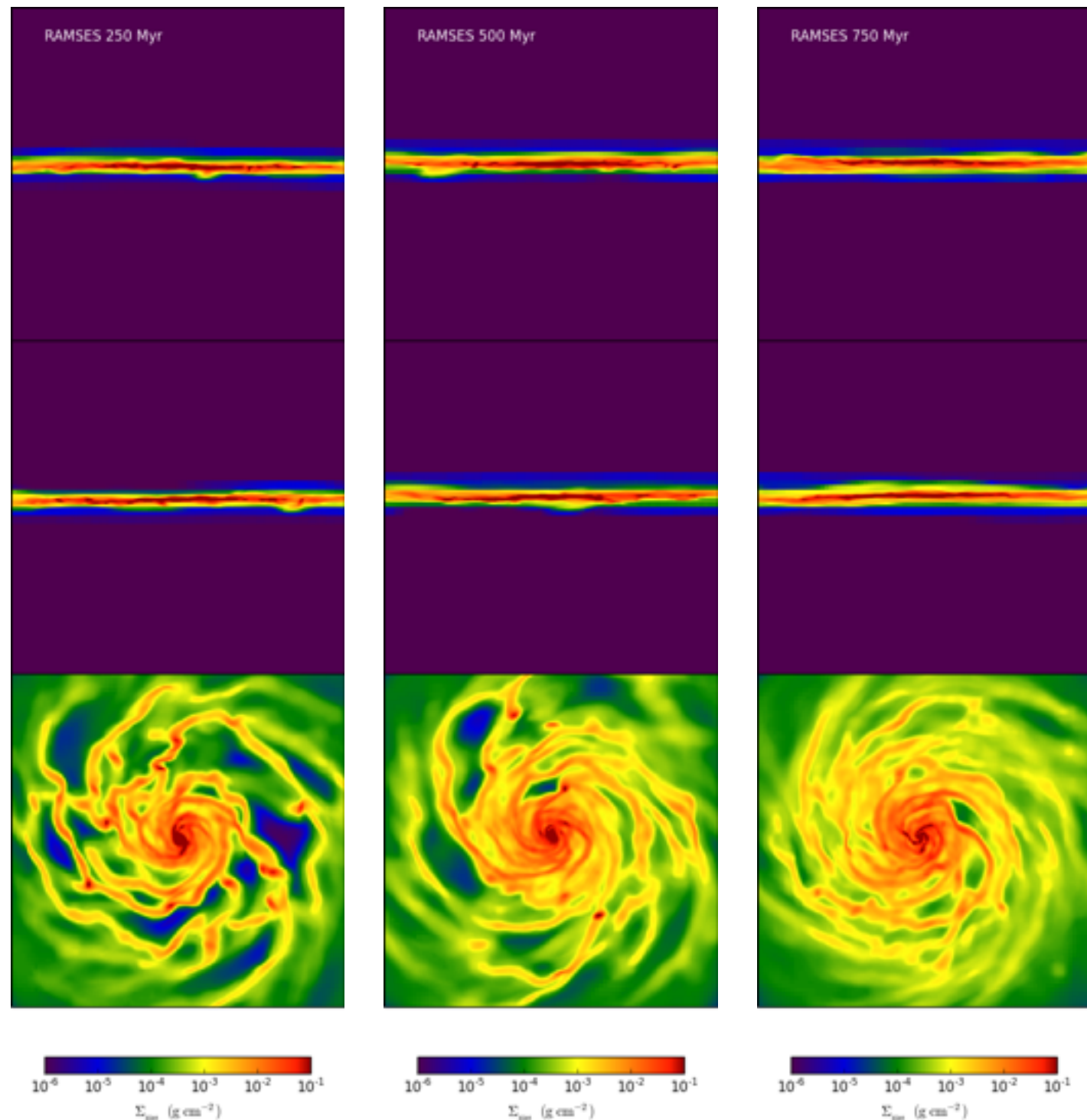
$$Z_{\text{gas}} = 1 Z_{\odot}$$



RAMSES EXAMPLES

Low res, cooling + star formation (no feedback)

- **YT visualization scripts**
(credit: Teyssier, Todoroki, Turk, Kim, etc.)

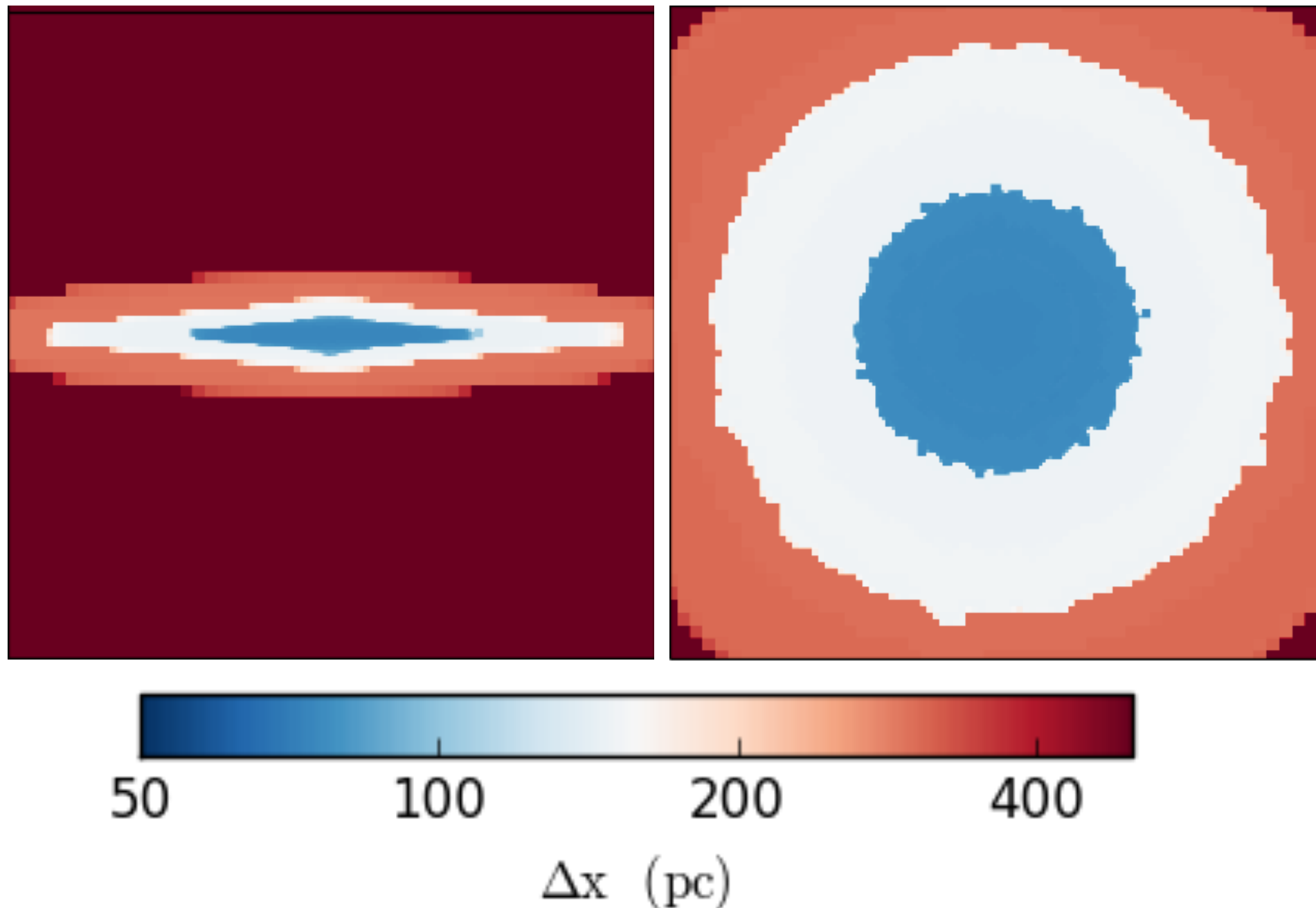


<https://bitbucket.org/mornkr/agora-analysis-script/src>

RAMSES EXAMPLES

Cooling + star formation (no feedback)

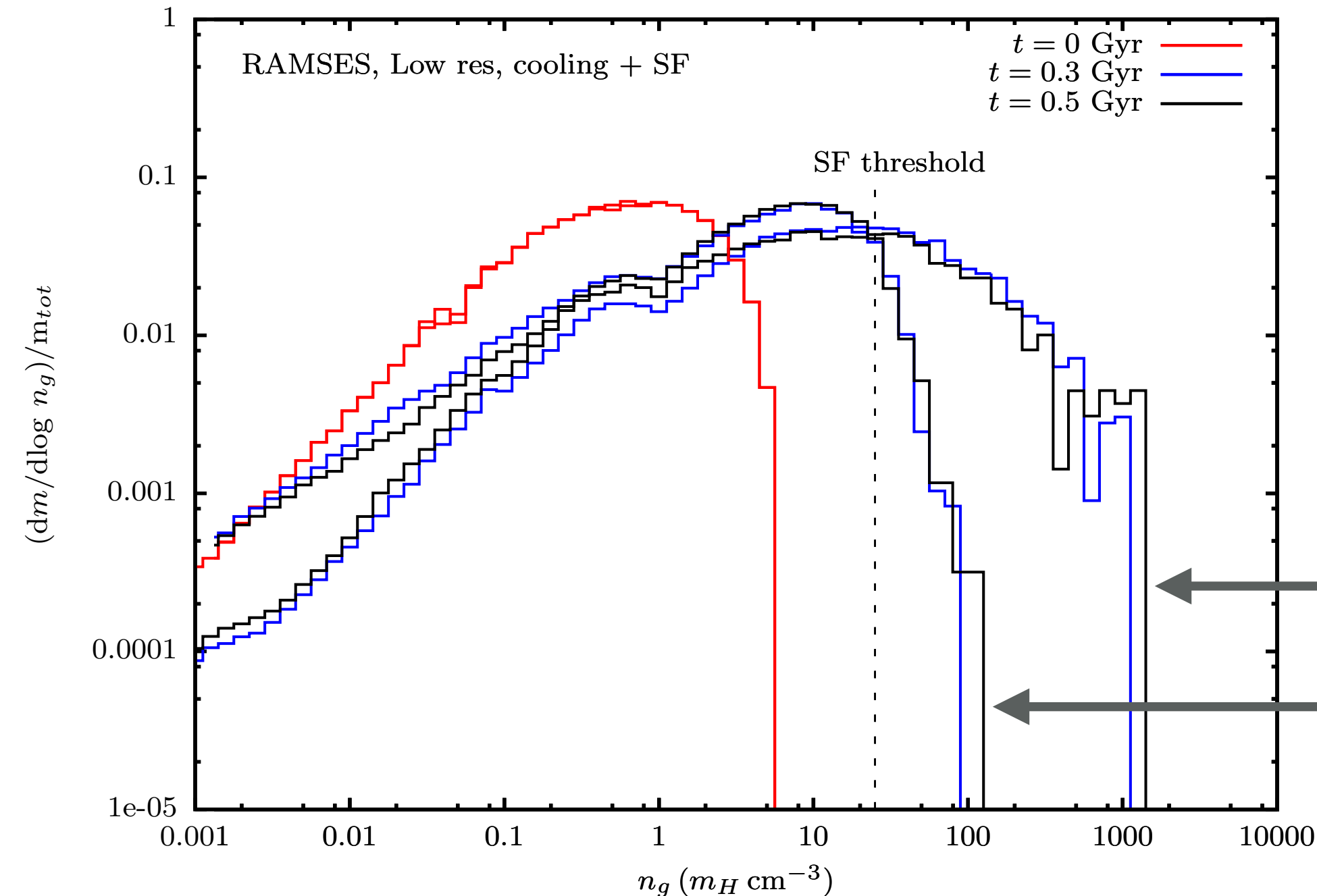
- Visualizing the numerical resolution/refinement strategy
(credit: Sam Leitner)



RAMSES EXAMPLES

Cooling + star formation

Mass weighted density PDF



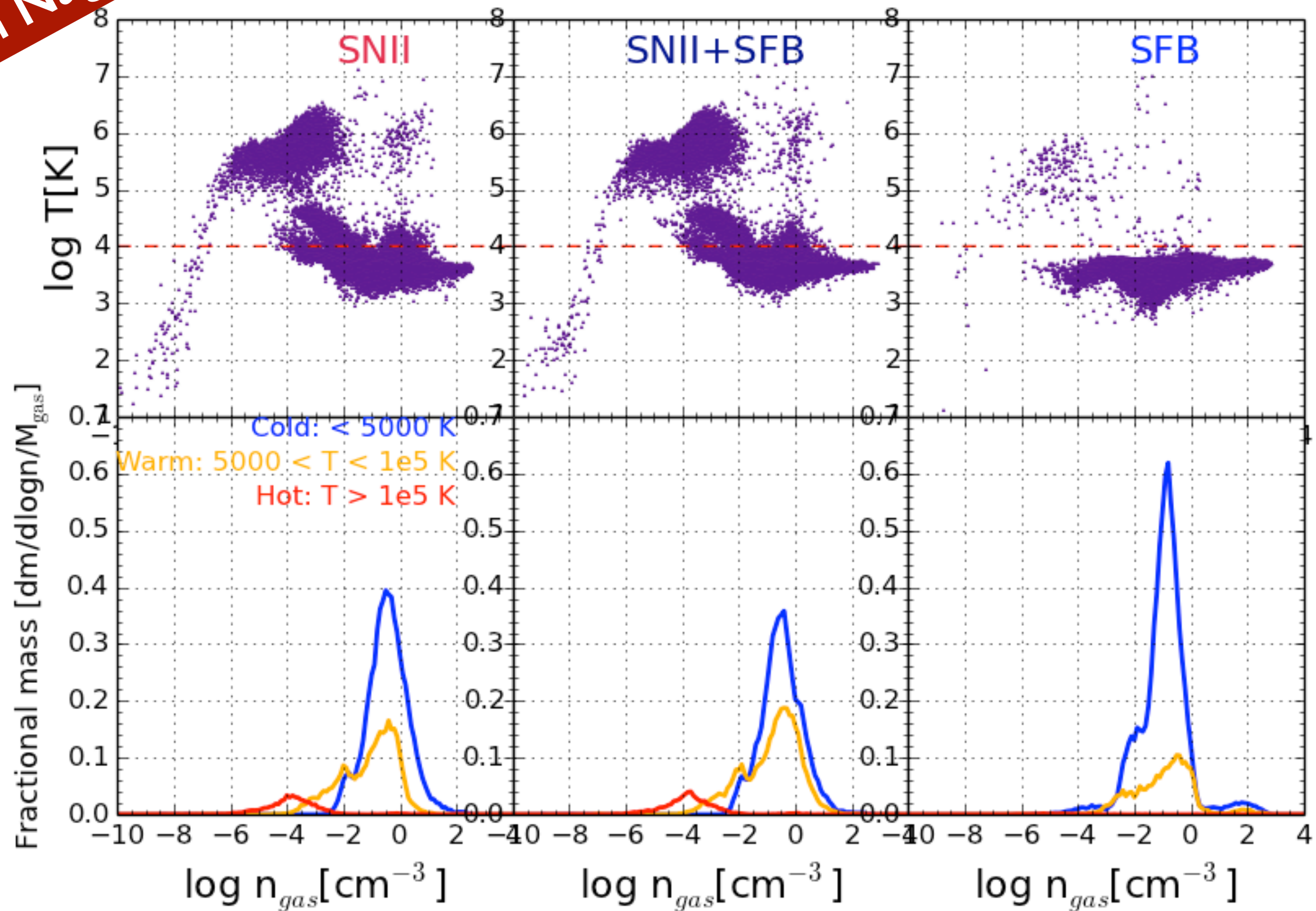
- Do we resolve the same density structure?
- How does (your pet recipe for) feedback affect the density PDF?

No feedback

Feedback

Gadget-3 example
Ken Nagamine's talk

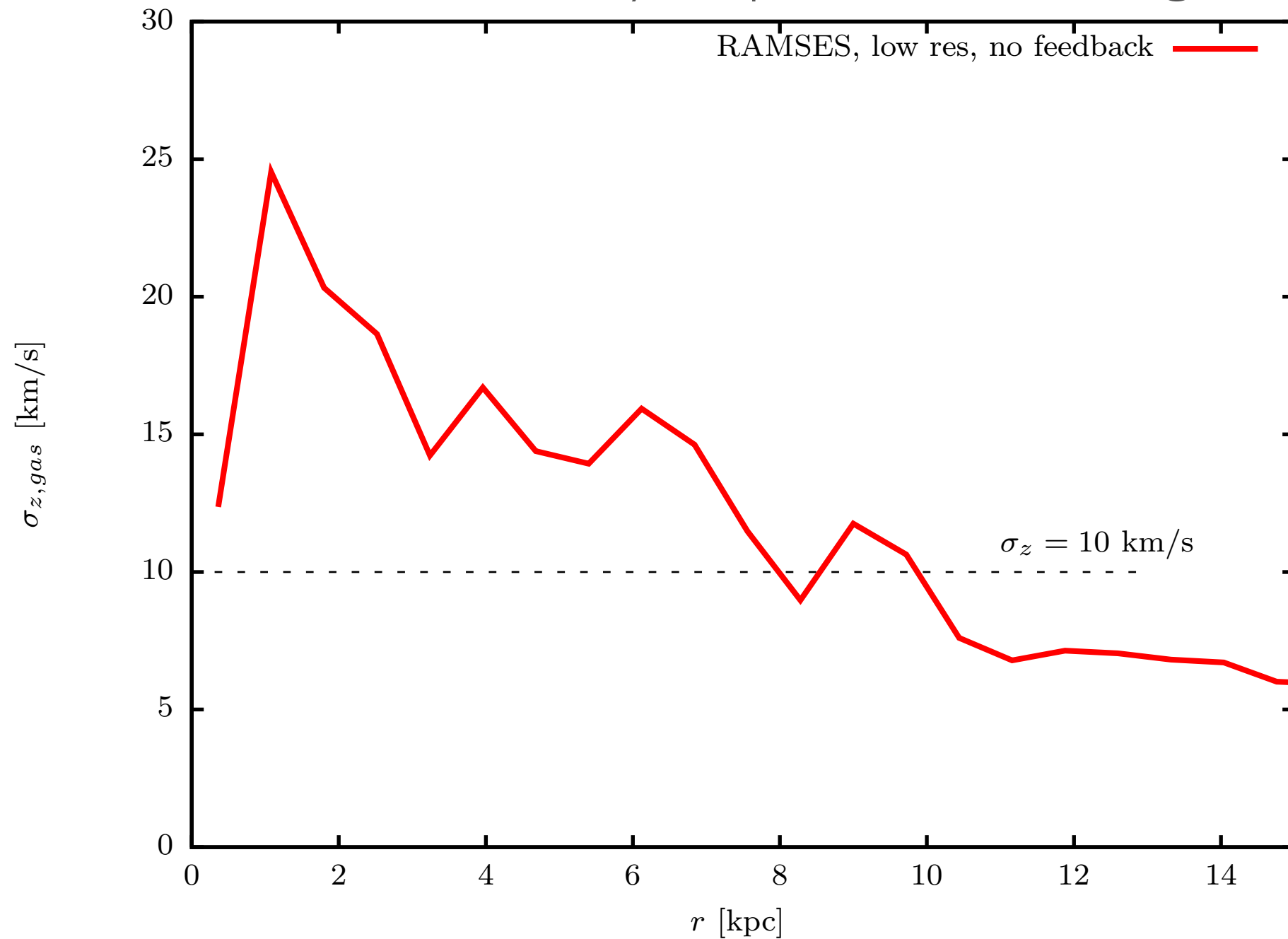
Phase Diagram



RAMSES EXAMPLES

Cooling + star formation (no feedback)

Vertical velocity dispersion in cold gas

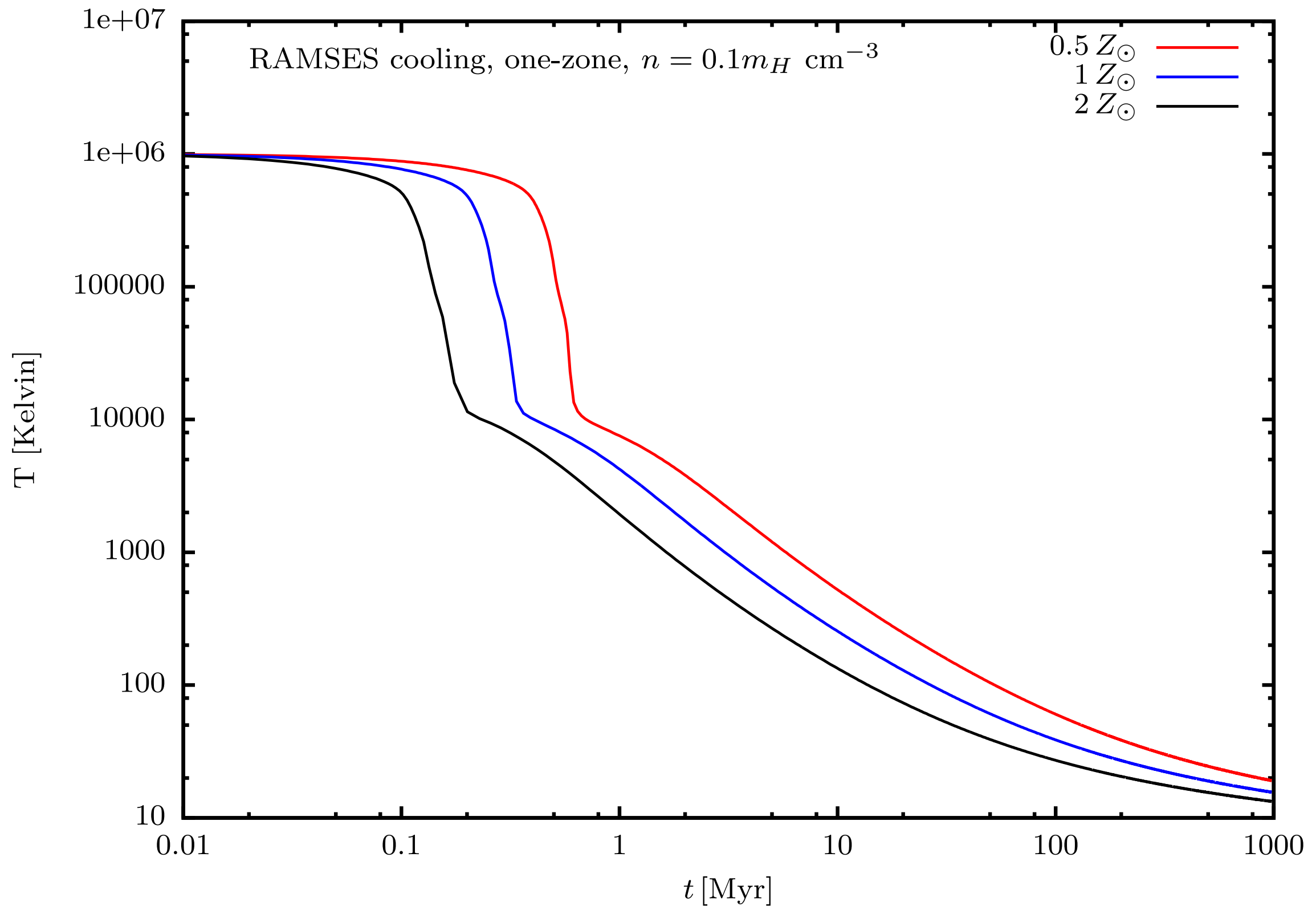


DISCUSSION, 16/8-2014

- **New Grackle API (Britton):** if the wrapper is in place for most of the codes, perform a comparison using a one-zone cooling test. An example of this would be to plot temperature vs time for a single cell starting at $n=0.1 \text{ cm}^{-3}$ and $T=10^6 \text{ K}$
- **Disk initial conditions:** compare isolated disk ICs (v_{rot} , Σ_{gas} , etc.)
- **Other subgrid physics:** Star formation, enrichment (Kim et al. 2014) and the use of a pressure floor. Should we all use the same star formation law, but allow for different way of sampling it?
- **Feedback:** agree on a baseline implementation of feedback for the disk test. If the cosmological runs are done with different feedback prescriptions, it may enough to do a simple thermal feedback injection here + a favorite FB model
- **Refinement strategy:** for mass based refinement in grid codes, what is the appropriate mass to refine on ($M_{\text{SPH kernel}}$, $M_{\text{gas particle?}}$). Choices for gravitational softening vs smoothing kernel size in SPH?
- **Actual comparison:** Agree on a suite of simulations for the isolated disk (e.g. cooling only, cooling+star formation, cooling+SF+feedback)
- **Analysis shopping list /suggestion for yt:**
 - Density maps of stars and gas (see [here](#))
 - Star formation histories (see [here](#))
 - Gas surface density, and star formation surface density, profiles
 - Velocity dispersion of cold gas
 - Density and temperature PDF (crucial to understand phase structure?) ([RAMSES example](#))
 - T-rho phase plots
 - Outflows: mass flux vs height over disk
 -

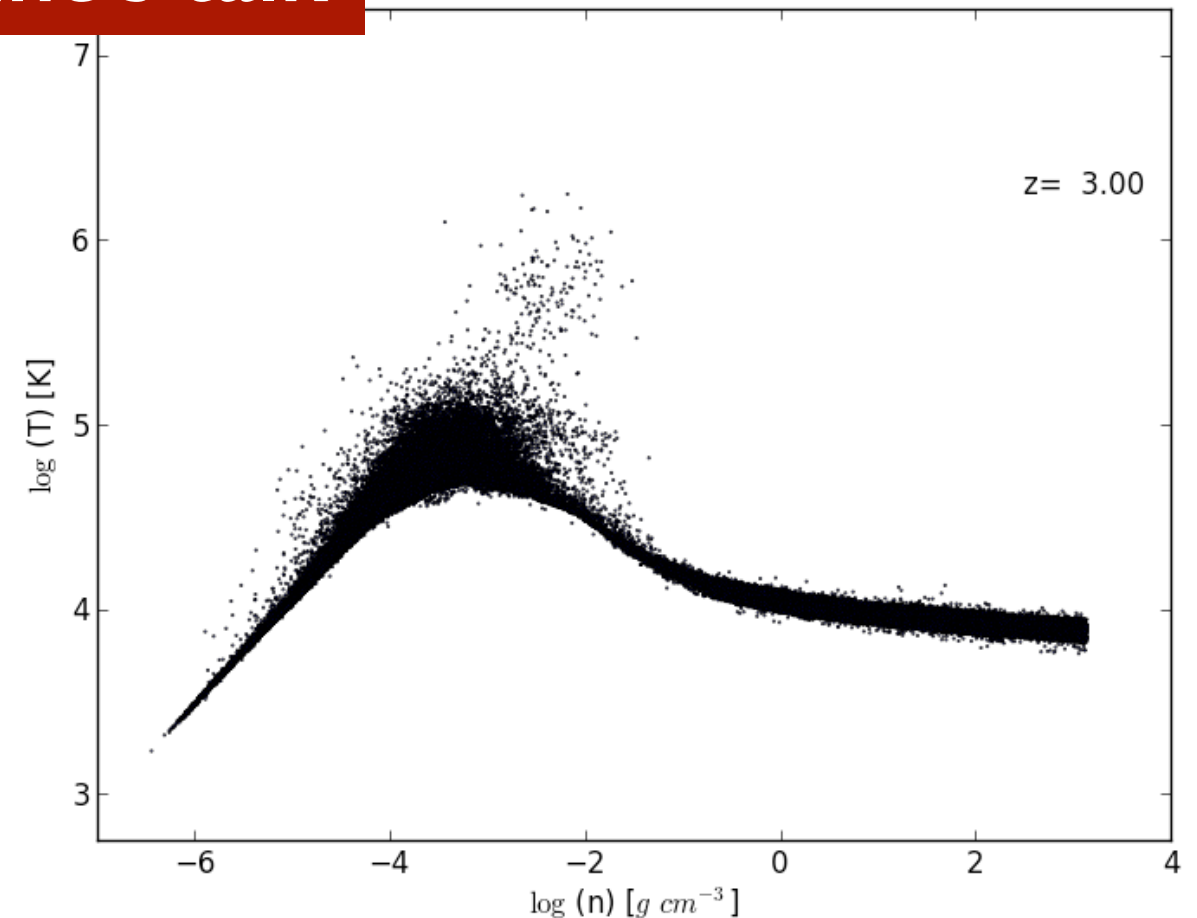
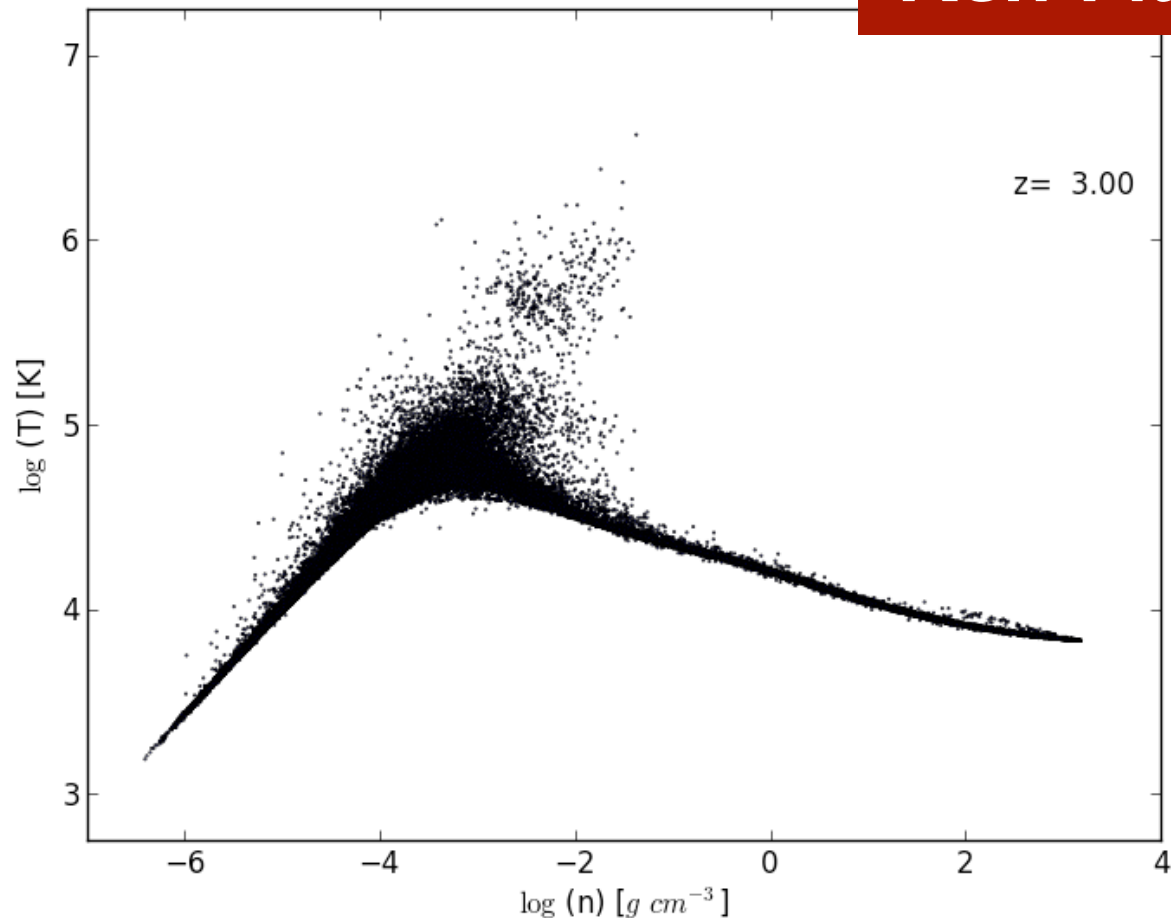
TEST OF GRACKLE IMPLEMENTATION?

One-zone cooling test



Comparing with original gadget cooling

Gadget-3 example
Ken Nagamine's talk



Gadget Cooling (Katz+ '96; Springel '05)
UVB : FG2011

Grackle cooling (i.e. CLOUDY)
UVB : FG2011

Overall agreement; seems to be working.

SUBGRID PHYSICS & SIMULATION SUITE

- **Star formation:** Schmidt law with simple density threshold $\dot{\rho}_* = \epsilon_{\text{ff}} \frac{\rho_{\text{gas}}}{t_{\text{ff}}}$
- **IMF & enrichment:** Kim et al. (2014)
- **Feedback:** For the cosmological runs, have we agreed on everybody using their own recipes?
- **Feedback for the isolated disk runs:** Increase complexity by
 - 1) cooling only
 - 2) cooling + SF
 - 3) cooling + SF + common simple feedback (up for debate! e.g. thermal SNe)
 - 4) cooling + SF + favorite feedback

Gas & Metal Recycling

Following the AGORA prescription (Kim+ '14):

Recycling fraction:

$$R = \int_{8M_{\odot}}^{40M_{\odot}} (m - w_m) \phi(m) dm \quad \text{with} \quad \phi(m) = dn/dm \propto m^{-2.3}$$

for a Chabrier IMF

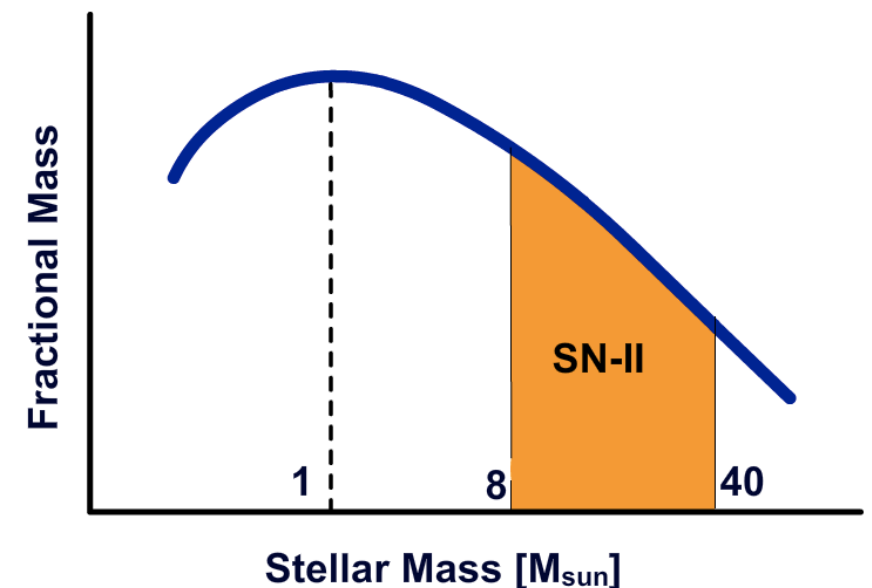
where $w_m = 1.4 M_{\odot}$ that is left behind as a remnant.

Metal production by SN-II

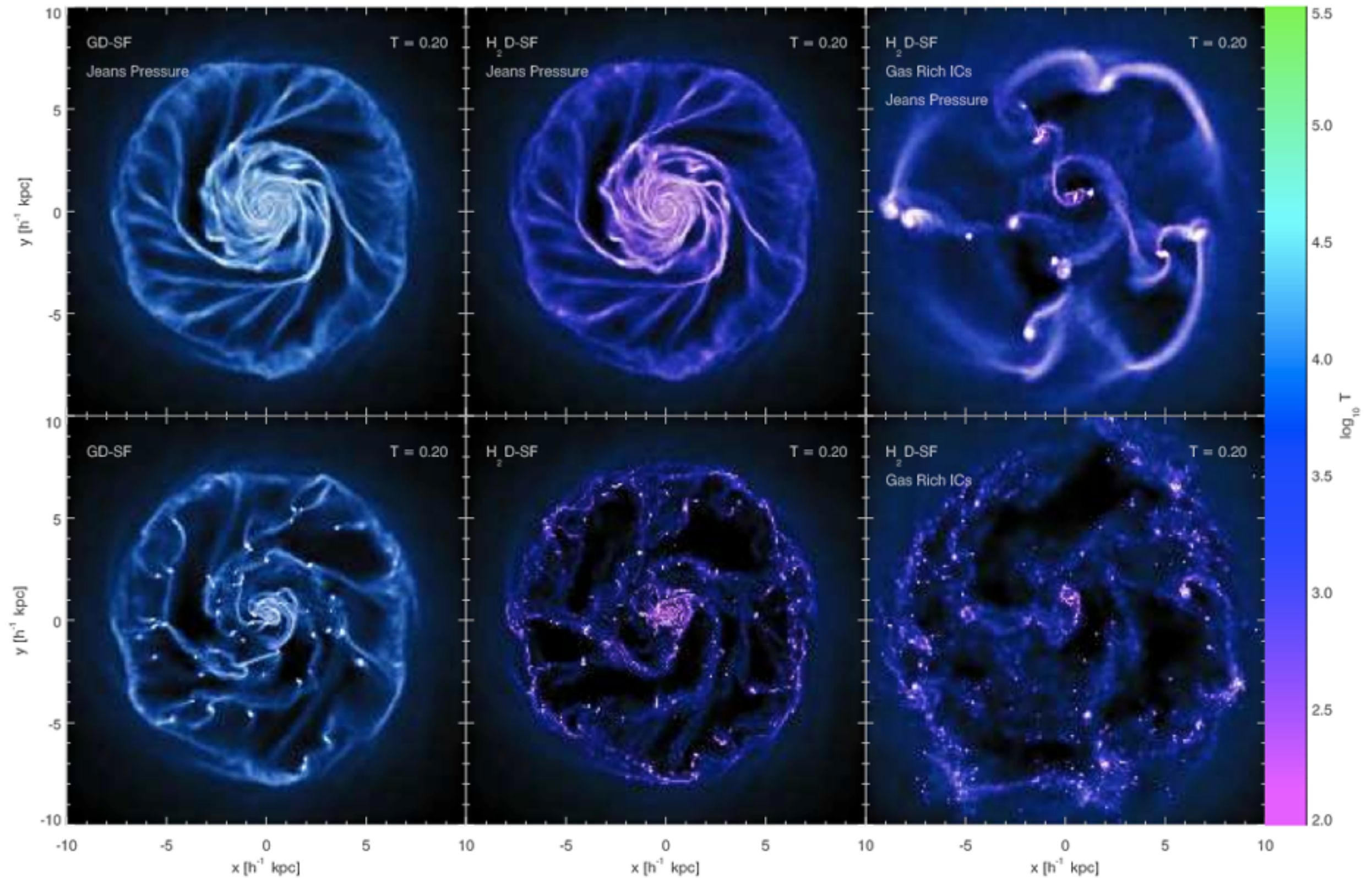
$$M_Z = 2.09M_O + 1.06M_{Fe}$$

(Kim et al. 2014)

→ Corresponding fractional masses: $0.0133 / M_{\odot}$ (O) and $0.011 / M_{\odot}$ (Fe)



(JEANS) PRESSURE FLOOR



Robertson & Kravtsov (2008)

(JEANS) PRESSURE FLOOR

- In AMR: resolve at least 4 Jeans lengths (e.g. Truelove et al. 1997):

$$\lambda_J = \sqrt{\frac{\pi c_s^2}{G\rho}}$$

$$\lambda_J = N_{\text{Jeans}} \Delta x$$

$$c_s^2 = \pi G \rho N_{\text{Jeans}}^2 \Delta x^2$$

- Strong fragmentation is less of a problem when turbulent pressure is involved. If the hydro kernel size is allowed to be much smaller than the force softening length, as is done in most modern cosmological SPH sims (e.g. Gasoline team, FIRE team), this may not be a problem: pressure gradients may support collapse.

- **As this topic is under debate, should we have runs both with and without a pressure floor!**

(JEANS) PRESSURE FLOOR AND REFINEMENT STRATEGY

README.TXT

Suggested run parameters and Jeans support

For all codes, the gas mass resolution has to be as close as possible to, in the case of the low resolution run, $8.593e4$ Msun. Using a Jeans mass 64x larger and the corresponding Jeans length, this results in a spatial resolution of 80 pc. We suggest that all codes, if possible, use this spatial resolution, or as close to it as possible. We also suggest, when applicable, to use a polytropic pressure floor, so that the Jeans length is resolved at all time. The polytrope is defined as $T=T_J (n_H/n_J)$ where $T_J=1800$ K and $n_J=8$ H/cc, for this resolution (LOW). We also suggest to use cooling at solar metallicity, with both atomic and fine structure (low T) cooling processes, if possible.

Table summarizing the suggested runtime parameters for the 3 resolution setting:

LOW:	$N_{\text{gas}}=1d5$, $m_{\text{gas_res}}=8.593d4$ msun, $\Delta x=80$ pc, $T_J=1800$ K, $n_J=8$ H/cc
MED:	$N_{\text{gas}}=1d6$, $m_{\text{gas_res}}=8.593d3$ msun, $\Delta x=20$ pc, $T_J=700$ K, $n_J=50$ H/cc
HIGH:	$N_{\text{gas}}=1d7$, $m_{\text{gas_res}}=8.593d2$ msun, $\Delta x=5$ pc, $T_J=300$ K, $n_J=300$ H/cc

What we have recommended to AMR users, is that we want refinement to be done in a Lagrangian fashion similar to SPH. What we mean by this is that the fundamental mass to refine on is $8 \times m_{\text{gas_res}}$ (i.e. a cell is split into an Oct, 8 cells) where $m_{\text{gas_res}}$ is the gas SPH particle mass for the different runs. This is a good starting point for AMR codes to attempt to do something similar to SPH methods. We continue to refine the grid down to the suggested resolution limit where the pressure floor kicks in. Once we get into details and start comparing our simulations we will study if a more suitable refinement criteria is necessary to match results.

- Refinement on m_{gas} or $8 \times m_{\text{gas}}$?

ANALYSIS SHOPPING LIST

- Density maps of stars and gas
- Star formation histories
- Disk structure: Gas surface density, and star formation surface density, profiles Vertical structure of various components.
- Velocity dispersion of cold gas
- Density and temperature PDF (crucial to understand phase structure)
- T-rho phase plots
- Outflows/inflow: mass/metal flux (spherical/cylindrical symmetry).
- Kennicutt-Schmidt relation
- Self-gravitating clumps (mass spectrum etc).
- Mass, momentum, entropy and energy conservation/evolution
- Disk stability analysis (e.g. Q_{gas} , Q_{stars} + multicomponent)
- Bulge-to-disk decomposition.
- V/σ